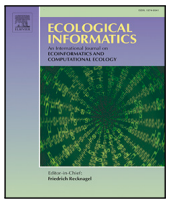
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The effect of collinearity between observed and model derived training variables on estuarine algal species distribution models

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ABSTRACT

Forecasts of organism distributions in time and space are needed to mitigate risks associated with changes in environmental conditions. These forecasts are often generated using correlative species distribution models (SDMs) that relate environmental variables to species presence or abundance. Biological complexity makes the construction of SDMs challenging because the collinearity between the environmental variables used to train the SDM may increase model parameter uncertainty. To analyze the effect of collinearity on SDMs, we (1) train SDMs for seven estuarine algal species commonly observed in the Chesapeake Bay (U.S.A.) using different levels of collinearity in the training information, (2) identify the environmental predictors, and (3) study their association with species presence using two statistical techniques (generalized linear models and regression trees). The novelty of our contribution is that our analysis uses both environmental *in situ* observations and environmental information generated by a mechanistic model. The environmental variables show strong collinearities in both the *in situ* observations (32 out of the total of 165 correlations) and mechanistic model output (12 out of the total of 120 correlations). To determine how collinearity between these variables affect our SDM results, we remove environmental variables that surpass a specific correlation threshold. We find that using these two different types of training information (i.e., observed vs. modeled) affects (1) the optimal set of predictors, (2) the associations between environmental variables and algal presence, and (3) the model's predictive skill. Water temperature is generally selected as an important predictor. Strong positive or negative associations between environmental variables and algal presence are not substantially impacted by the type of training information used. Although removing collinearities may result in the detection of new important predictors, it may also result in a slight decrease (~ 5 %) of the SDM prediction skill, depending on the species of interest and type of training information. Our findings suggest that the main environmental predictors rely on both the species characteristics and training information type used in SDM construction, and highlight the challenge of interpreting the associations between environmental conditions and species presence predicted by these SDMs. These insights help us to better understand the environmental conditions of importance to these algal species and hence optimize monitoring efforts by revealing which *in situ* observations are vital to accurately forecast blooms of these estuarine algae.

1. Introduction

Correlative species distribution models (SDMs) have been broadly used to predict species presence and their shifts in time and space. They have been successfully applied to predict both terrestrial (Elith, 2000; Elith et al., 2010; Van Doren and Horton, 2018) and aquatic species distributions (Anderson et al., 2010; Fabrizio et al., 2021; Horemans et al., 2023a; Silva et al., 2024; Woo et al., 2025; Brandenburg et al., 2025). Correlative SDMs are often preferred over mechanistic models for

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predicting species presence because the latter require presumptions and significant information regarding growth and mortality rates (Franks, 2002; Ralston and Moore, 2020), which are challenging to determine given the complexity of species dynamics (Franks, 2009; Friedrichs et al., 2006; Horemans et al., 2023a). Another advantage of using correlative SDMs is their ability to relate environmental conditions to organism presence without the need for *a priori* insight into the species dynamics.

Interpreting the associations between environmental variables and species presence using correlative SDMs is challenging. A first challenge is related to the intrinsic biological complexity of species dynamics. To illustrate this, we examine the so-called environmental 'niche' concept (Hutchinson, 1957). Pulliam (2000) classified the relationship between environmental niche and species distribution into four categories: (a) the Grinnellian Niche, in which a species arises whenever the environmental conditions are suitable and nowhere else, (b) the Hutchinsonian Realized Niche, in which species will be absent in some parts of the niche space that are dominated by other competitors, (c) Source-Sink Dynamics, in which a species can only persist because of immigration from other source habitats, and (d) Dispersal Limitation, where species are absent from suitable habitats because of frequent extinction events. Connecting the associations between environmental variables and species presence highly depends on which niche category (or combination) is applicable (see also Section 4.4 for further discussion of this concept).

A second challenge of relating environmental conditions to species presence is the choice of training information (Elith et al., 2010). Environmental variables in the training information are commonly correlated (Dormann et al., 2013; Ralston and Moore, 2020). This collinearity increases uncertainty in the fitted SDM parameters that may hinder identification of the main environmental predictors and impede the detection of important associations between environmental variables and organism presence (Dormann et al., 2013). Additionally, correlative SDMs are commonly trained using *in situ* observations yet applied using environmental mechanistic model outputs when generating forecasts (Brown et al., 2013; Baker et al., 2018; Van Doren and Horton, 2018; Ralston and Moore, 2020). The difference in information type used to train versus apply the correlative SDM may affect the applicability and interpretation of the correlative SDM because of the difference between mechanistic model outputs and *in situ* observations (Horemans et al., 2024).

Various authors have studied the effect of correlative SDM types (e.g., generalized linear models and random forest) on species distribution forecasting (Elith, 2000; Pearson et al., 2006; Anderson et al., 2010; Valavi et al., 2022; Szewczyk et al., 2025). However, little is known of the impact of different types of training information on the construction and interpretation of correlative SDMs (e.g., Elith et al., 2010; Dormann et al., 2013; Fabrizio et al., 2021; Horemans et al., 2024). Our contribution focuses on answering how both the level of collinearity between environmental variables and the use of environmental *in situ* observations versus mechanistic model output to train the correlative SDM affect the selection of the main environmental predictors, their associations with species presence, and SDM prediction skill. Although we create correlative SDMs for seven estuarine taxa that are often found in the Chesapeake Bay (U.S.A.) using two standard methods – generalized linear models and regression trees – our primary emphasis is on generalized linear models. We hypothesize that the type of training information (i.e., with varying levels of collinearity and environmental *in situ* versus mechanistic model output) determines the selection of the main environmental variables in the SDM and their association with algal bloom presence and our skill to predict their occurrence.

2. Methods

2.1. Statistical model experiments

We train correlative SDMs using different types of information, that is, with varying levels of collinearity and environmental *in situ* observations versus mechanistic model output. We determine how these various types of training information affect the selection of the main environmental predictors, their associations to species presence, and the SDM prediction skill (Fig. 1). SDMs are constructed for seven estuarine algal taxa (mostly harmful) that regularly bloom in the Chesapeake Bay in various regions and seasons (Table 1; Horemans et al., 2024). The environmental mechanistic model output is generated using a 3D hydrodynamic-biogeochemical model (Bever et al., 2021; St-Laurent and Friedrichs, 2024). We compute the Pearson correlation coefficient ρ between the environmental training variables X_i and Y_i to quantify collinearity using both the environmental *in situ* observations and mechanistic model output:

$$\rho = \frac{\mathbb{E}[(X_i - \overline{X})(Y_i - \overline{Y})]}{\{\mathbb{E}[(X_i - \overline{X})^2]\mathbb{E}[(Y_i - \overline{Y})^2]\}^{1/2}} \quad (1)$$

in which the $\mathbb{E}$ and overline denote the expected value and mean, respectively. We consider four correlation thresholds when adjusting the level of collinearity $|\rho| > 0.4$, 0.5, 0.6, and 0.7. When removing collinearity, we prioritize removing biogeochemical training variables over physical ones because they have higher observational and model uncertainties. Finally, we compare the main environmental predictors and corresponding prediction skill of these various SDMs constructed using different types of training information. In the following sections, we introduce the environmental *in situ* observations and mechanistic model output, correlative model type, and metrics for SDM prediction skill. For additional details, we refer the reader to the Overview, Data, Model, Assessment, and Prediction (ODMAP) reporting (Zurell et al., 2020) in the Supplementary Material [adapted from Horemans et al. (2024, 2023b)].

2.2. Training information

2.2.1. Environmental in situ data

The *in situ* observations cover 1985–2020 and various depths at 42 stations distributed over both the tributaries and main channel of the Chesapeake Bay (publicly accessible through the online database: Chesapeake Bay Program, 2025). Algal cell count data (total of 7,059 per taxon) are transformed to bloom data by assuming a fixed cell count bloom threshold for each taxon based on the local literature (Table 1). We refer the reader to the Supplementary Material for the resulting spatial and temporal distribution of the observed blooms (Figs. S1–S2). Algal bloom observations are connected to corresponding environmental conditions based on depth, location, and day. We focus on nineteen environmental variables that are known to affect bloom presence (Table 2; Wells et al., 2015; Horemans et al., 2023b). These include both physical and biogeochemical variables. Apparent oxygen utilization, defined as the oxygen concentration at saturation minus the *in situ* observed oxygen concentration, is estimated from dissolved oxygen, temperature, and salinity (U.S. Geological Survey, 2011) because this variable is a better measure of biological activity than absolute oxygen concentration, which includes impacts due to solubility. The vertical gradient in temperature and salinity are calculated by dividing the differences of the averaged values below and above the pycnocline (i.e., depth at which we observed a density discontinuity) by the averaged distance between the observations. Solar irradiance, wind, and precipitation data are obtained from the ECMWF Reanalysis v5 (ERA5) reanalysis (Hersbach et al., 2020). The estimates of solar irradiance are translated to mean Bay-wide solar irradiance dependent solely on Julian day. Multi-annual averages of water depth are computed at each station.

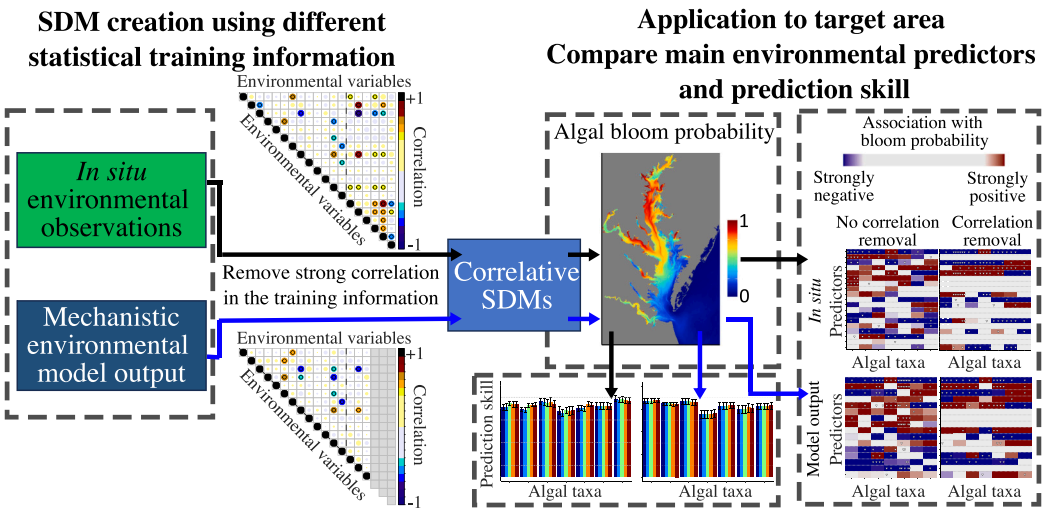


Fig. 1. A schematic overview of our approach to study the impact of statistical training information used to construct species distribution models (SDMs) on the environmental predictors, their association with species presence, and prediction skill. See Section 2 for additional information on the study's approach.

Table 1
The seven estuarine algal taxa (mostly harmful) for which we construct species distribution models and their corresponding bloom threshold value and number of blooms in our data set.
Source: Table adapted from Horemans et al. (2024).

Taxon name	Number of blooms	Bloom threshold value	References
Prorocentrum minimum (or *P. cordatum*)	360	1,000 cells mL^{-1}	Marshall and Egerton (2009), Pease et al. (2021), and Mulholland et al. (2018)
Heterocapsa rotundata	617	1,000 cells mL^{-1}	Marshall and Egerton (2009) and Mulholland et al. (2018)
Heterocapsa triquetra (or *steinii*)	201	200 cells mL^{-1}	Baek et al. (2011) and Marshall and Egerton (2009)
Microcystis sp.	260	10,000 cells mL^{-1}	Marshall and Egerton (2009) and Ho and Michalak (2015)
Pseudo-nitzschia pungens	261	500 cells mL^{-1}	Anderson et al. (2010)
Pseudo-nitzschia seriata	163	250 cells mL^{-1}	Anderson et al. (2010)
Dinophysis acuminata	191	0.4 cells mL^{-1}	Díaz et al. (2016)

2.2.2. Environmental mechanistic model output

We pair each *in situ* observation with mechanistic environmental model output based on depth, location, and day. The model-derived environmental fields are available from the Chesapeake Bay Environmental Forecast System (CBEFS) (www.vims.edu/cbefs Bever et al., 2021), which is an implementation of the Regional Ocean Modeling System (Shchepetkin and McWilliams, 2005) for the Chesapeake Bay (ROMS-ECB; Feng et al., 2015; Frankel et al., 2022; St-Laurent and Friedrichs, 2024). This model has been described in detail and evaluated against independent observational data sets in multiple publications (Turner et al., 2021; Frankel et al., 2022; Hinson et al., 2023; St-Laurent and Friedrichs, 2024) (also see Supplementary Material for an additional comparison of model outputs with observational data; Fig. S3).

Because not all *in situ* environmental variables are modeled, we focus on only sixteen modeled environmental variables instead of all nineteen present in the *in situ* observations (Table 2). We also add three new, model-derived variables to our set of environmental predictors: vertical mixing and exponential light extinction coefficient, both of which affect the light availability, and residual water flow velocity, which determines whether algal concentrations can build up or are flushed out (cf., residence time). Given that the response timescale of an algal bloom is typically $\gtrsim \mathcal{O}$(1 day) (Fan and Glibert, 2005), we use daily averaged values from the mechanistic model. The exception is the residual water flow velocity, which is computed as the 14-day averaged water flow (before the observation of the algal cell count) projected in the direction of the long-term, 355-day residual water flow. We choose 14 and 355 days to address the effect of tides (14 days ≈ 1 neap-spring cycles and 27 M_2 cycles; 355 days ≈ 24 neap-spring cycles, 686 M_2 cycles, and roughly one year). This variable can be either positive (14-day average is pointing in the direction of the long-term residual water flow) or negative (14-day average is pointing in the opposite direction of the long-term residual water flow).

2.3. Correlative species distribution models

Although the SDMs of the seven algal taxa are constructed based on two standard methods – generalized linear models (GLMs) and regression trees – our main focus is on GLMs as their results are sufficient to support our main findings. However, the results from these GLMs are later compared against those of SDMs constructed using regression trees, which may capture nonlinear relationships between environmental variables and species presence (see the Supplementary Material for a methodological description of the regression trees). In the GLM analysis, it is assumed that the binary algal bloom data follow a binomial distribution and that logit η:

$$\eta = \ln\left(\frac{p}{1-p}\right), \tag{2}$$

Table 2
The environmental variables considered in the Chesapeake Bay species distribution models and their corresponding mean and standard deviation ($N = 7,059$; between 1985–2020) for both the *in situ* observations and mechanistic model output.

Environmental variable	Mean		Standard deviation		Units
	in situ	model	*in situ*	model	
Water temperature (T)	17.2	17.5	7.9	7.4	°C
Total organic nitrogen (TON)	0.46	0.40	0.20	0.14	mg L^{-1}
Salinity (S)	16.0	15.2	8.1	8.3	/
Apparent oxygen utilization (AOU)	0.81	0.34	1.9	1.8	mg L^{-1}
Total water depth	16.3	16.3	6.7	6.7	m
pH	7.9	8.1	0.37	0.36	/
Solar irradiance at the water surface (swrad)[a]	188	188	55	55	W m^{-2}
Dissolved inorganic nitrogen (DIN)	0.23	0.37	0.35	0.52	mg L^{-1}
Vertical gradient of salinity (gradS)	0.38	0.40	0.33	0.30	m^{-1}
Precipitation (rain)[a]	1.6		4.0		mm day^{-1}
Magnitude of the wind velocity (wind)[a]	3.3		1.6		m s^{-1}
Ammonium concentration (NH_4)	4.8×10^{-2}	8.3×10^{-2}	6.3×10^{-2}	9.9×10^{-2}	mg L^{-1}
Vertical gradient of the water temperature (gradT)	−0.10	−0.12	0.20	0.24	°C m^{-1}
Total dissolved phosphorous concentration (TDP)	1.8×10^{-2}	/	1.5×10^{-2}	/	mg L^{-1}
Total nitrogen concentration (TN)	0.69	/	0.43	/	mg L^{-1}
Total suspended solids concentration (TSS)	13.6	/	13.3	/	mg L^{-1}
Silica concentration (Si)	0.74	/	0.75	/	mg L^{-1}
Total phosphorous concentration (TP)	4.3×10^{-2}	/	3.0×10^{-2}	/	mg L^{-1}
Ratio of the total dissolved nitrogen to phosphorous concentration (N:P)	19.1	/	11.5	/	/
Vertical mixing coefficient (D)	/	6.3×10^{-4}	/	9.2×10^{-4}	m^2 s^{-1}
Exponential light extinction coefficient (k_d)	/	0.95	/	0.47	m^{-1}
Residual water flow velocity (flow)	/	2.8×10^{-2}	/	6.3×10^{-2}	m s^{-1}

[a] Deduced from the ERA5 reanalysis (Hersbach et al., 2020).

in which n is the total number of trials and p is the probability of a bloom, can be approximated by a linear combination of the environmental variables x_i:

$$\eta = \beta_0 + \sum_{i=1}^{m} \beta_i x_i \tag{3}$$

in which m is the number of environmental variables included in the GLM, $\beta_0, \ldots, \beta_m$ are the corresponding unknown model parameters that are fitted by maximizing a likelihood function, and variables x_i are standardized:

$$x_i = \frac{\tilde{x}_i - \text{mean}(\tilde{x}_i)}{\text{std}(\tilde{x}_i)}, \tag{4}$$

in which $\tilde{x}_i$ is the dimensional form of x_i with corresponding mean and standard deviation mean($\tilde{x}_i$) and std($\tilde{x}_i$), respectively. The advantage of using GLMs is that once we have estimated $\beta_0, \ldots, \beta_m$, we can compute the probability of bloom occurrence p given a set of environmental variables using Eq. (2). Furthermore, GLMs allow us to easily determine the optimal choice of environmental variables to forecast blooms and the association between these variables and bloom presence, which is quantified by the sign (positive or negatively associated to bloom probability) and magnitude (greater magnitude, more important) of the β-value corresponding to each environmental variable. Following Horemans et al. (2023b), we train 50 GLMs since the random selection of a training and test data set affects the estimation of the β-values. We compute the averaged β-value signs of these 50 GLMs for each environmental variable:

$$\beta_i^{\pm} = \frac{1}{50} \sum_{k=1}^{50} \text{sign}(\beta_i^k), \tag{5}$$

in which β_i^k represents the β-value corresponding to the kth GLM (out of a total of 50) and environmental variable x_i. Sign denotes the sign function (i.e., +1, 0, or −1). $\beta_i^{\pm}$ ranges between −1 (always associated with a decrease of bloom probability) and +1 (always associated with an increase of bloom probability) and is thus a measure of the consistency in whether environmental variable x_i is associated with an increase or decrease of bloom probability. To avoid overfitting the GLM, we only choose up to five environmental variables to train the GLMs and use the Akaike Information Criterion (Akaike, 1998) to select the optimal combination out of the total of 16,663 (i.e., up to 5 out of a set of 19 variables) or 4,943 (i.e., up to 5 out of a set of 15 variables) variable combinations using the *in situ* observations or mechanistic model outputs, respectively. If combinations are indistinguishable using a statistical two-tailed Welch t-test (p-value > 0.05), then we select multiple optimal combinations. The skill of the GLM is evaluated using the model accuracy in forecasting a bloom α_1 and no-bloom event α_0:

$$\alpha_1 = \frac{N_1^{pred}}{N_1^{total}}, \tag{6}$$

$$\alpha_0 = \frac{N_0^{pred}}{N_0^{total}}, \tag{7}$$

in which N_1^{pred}, N_0^{pred} are the number of correct GLM predictions and N_1^{total}, N_0^{total} the total number of bloom and no-bloom occurrences, respectively. To compute N_0^{pred} and N_1^{pred}, we assume that we detect a bloom when bloom probability $p > 50$ %, which is evaluated using the *in situ* algal cell counts.

3. Results

3.1. Collinearity analysis

Strong collinearities are evident between the environmental variables regardless of information type (Fig. 2; the actual values are listed in the Supplementary Material Fig. S4). Both the *in situ* observations and mechanistic model output show relatively strong positive collinearities (i.e., $\rho > +0.5$) between T and swrad, AOU and NH_4, and DIN and NH_4 and negative collinearities (i.e., < -0.5) between AOU and pH, S and DIN, and swrad and gradT. Focusing on the matching variables (cf., those to the left of the dashed line in Fig. 2), the main differences between the two data sets are the strong positive correlation between TON and pH in the mechanistic model output and the strong negative correlation between TON and S in the *in situ* observations.

3.2. Generalized linear models

3.2.1. Associations between environmental variables and blooms

In general, removing collinearity between variables does not affect strong associations between environmental variables and bloom presence, regardless of the type of information (Fig. 3). We only show

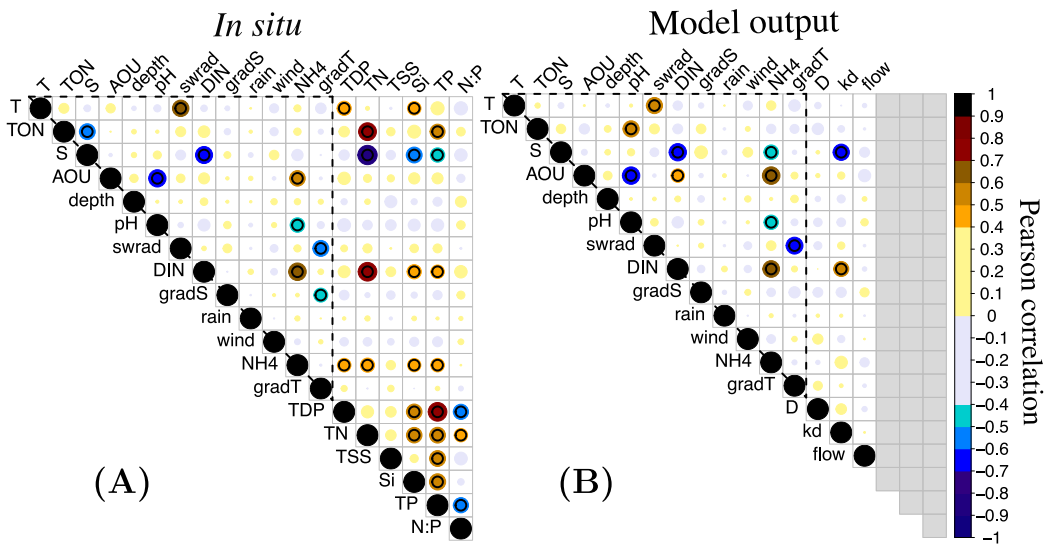


Fig. 2. Collinearity between the environmental variables as quantified by the Pearson correlation using (A) *in situ* observations and (B) mechanistic model output. The definition of the acronyms are listed in Table 2. Correlations for which the absolute value $\geqslant 0.4$ are denoted by black circles and variables to the left of the dashed line are the overlapping variables between the *in situ* observations and model output.

associations between environmental variables and bloom probability [as defined in Eq. (5)] using a correlation threshold of 1 (cf., no threshold; Figs. 3A and 3C) and 0.4 (Figs. 3B and 3D), though the main conclusions are the same when other threshold values are applied (not shown). However, the strength of an association may change when removing variables based on their correlation (Figs. 3C–D; crossed out rows). For example, a correlation threshold of 0.4 results in a stronger association between salinity and bloom probability when using *in situ* observations for *Heterocapsa triquetra* and *Pseudo-nitzschia seriata* (Fig. 3B), which are not found when no correlation threshold is applied (Fig. 3A). Overall, when strong associations are found using both the *in situ* observations and mechanistic model outputs (depicted in red and blue), they have the same sign. Exceptions do exist, such as the negative associations (cf., blue) found for TON and pH for *Pseudo-nitzschia seriata* using *in situ* observations, which are positive (cf., red) when using mechanistic model outputs. The associations and the selection of the optimal variable combination (depicted by the triangles) depend on the taxon of interest. Some taxa show multiple optimal sets that are statistically indistinguishable. For example, *Heterocapsa rotundata* has two optimal sets when the SDM is trained using *in situ* observations: {T, TON, gradT, TSS, Si} and {T, TON, swrad, TSS, Si}. The optimal variables also depend on the type of training information. With the exception of the *Dinophysis acuminata* SDM, water temperature is always selected as an important predictor of a bloom, independent of taxon or training information type.

3.2.2. Species distribution model skill

The accuracies of forecasting a bloom occurrence using *in situ* observations typically range between ~ 65%–90% and significantly decrease with decreasing correlation threshold for *Prorocentrum minimum*, *Heterocapsa rotundata*, *Microcystis sp.*, and *Pseudo-nitzschia pungens* and stay the same or improve for *Heterocapsa triquetra*, *Pseudo-nitzschia seriata*, and *Dinophysis accuminata* (p-value < 0.05 using a Welch two-tailed t-test; Fig. 4A). The accuracies of forecasting a no-bloom occurrence similarly range between ~ 65%–95% (Fig. 4B). Implementing the lowest correlation threshold (blue curve) results in a significant decrease of these accuracies compared to the case without correlation removal (red curve) for all algal taxa. In contrast, although the accuracies of forecasting a bloom when using mechanistic model outputs also range between ~ 65%–95% (Fig. 4C), these accuracies are not significantly affected by the implementation of a correlation threshold (Welch two-tailed t-test, p-value > 0.05; Fig. 4C). The accuracies of forecasting a no-bloom event, which range between ~ 70%–95% (Fig. 4D), however, do show a significant difference between the highest and lowest correlation threshold, except for *Heterocapsa triquetra* ($p = 0.50$) and *Pseudo-nitzschia seriata* ($p = 0.77$).

3.3. Regression trees

The results using the regression trees instead of GLMs show very similar patterns. In general, removing collinearity between variables does not impact strong associations between environmental variables and bloom presence (Fig. S5). However, as with the GLM, the strength of an association may change when excluding variables based on their correlation. Overall, strong associations have the same sign when detected using both *in situ* observations and mechanistic model outputs, and their signs consistently overlap when comparing the GLM and regression tree results (Fig. S5). As with the GLM results, water temperature remains the most important predictor of bloom occurrence. Regarding model skill (Fig. S6), although regression trees show slightly better accuracies (~ a few percentages) than GLMs, they also typically range between ~ 65%–95%. The regression tree analysis also shows some differences compared to the GLMs: we detect less variability in the selection of optimal combination of predictors and fewer strong associations using regression trees.

4. Discussion

4.1. The importance of training information

Training information (i.e., removing collinearity and training with *in situ* observations versus mechanistic model output) affects whether we can detect strong associations between environmental variables and algal bloom probability, and which of these variables are selected as optimal environmental predictors for blooms (Fig. 3). Additionally, the type of training information may alter forecasting accuracy (Fig. 4). These results agree with those of other studies. Dormann et al. (2013) showed that collinearity in the training information inflate the variance of regression parameters in correlative SDMs. In our applications, this variance will hinder the detection of strong associations between environmental variables and bloom probability. For instance, by removing *in situ* variables that are strongly correlated with salinity, a strong association was detected between salinity and bloom occurrence for *Heterocapsa triquetra* and *Pseudo-nitzischa seriata*, which was absent

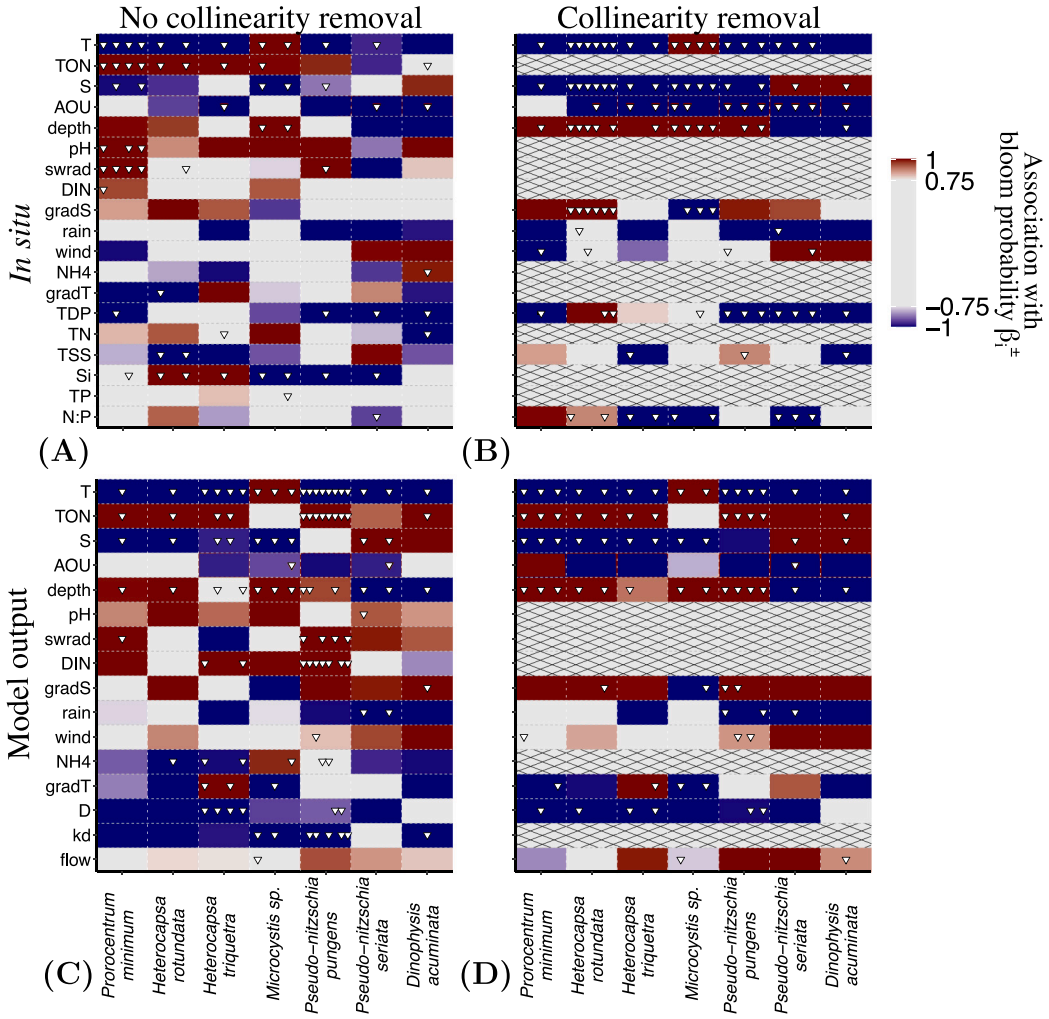


Fig. 3. Associations between environmental variables and algal bloom probability [see Eq. (5)], and the optimal variable combinations to forecast blooms (white triangle symbols inside the rectangles) for the seven algal taxa examined without (cf., correlation threshold of 1; left panels) and with (right panels) removal of collinearity using environmental (A)–(B) *in situ* observations and (C)–(D) mechanistic model output. The hashed rows depict the environmental variables that were removed based on a correlation threshold of 0.4. The definition of the acronyms are listed in Table 2.

when all variables were considered. In regards to model skill, Elith et al. (2010) constructed various correlative SDMs to forecast cane toad (i.e., *Bufo marinus*) presence in Australia. They found that not only model skill, but also optimal variable combinations are affected by adding environmental information (derived from a mechanistic model). Other examples, Fabrizio et al. (2021) and Horemans et al. (2024) improved their correlative SDM skill for forecasting four key forage fish and the algal taxa considered in this study, respectively, by using mechanistic model outputs instead of *in situ* observations as training information.

4.2. Associations between environmental variables and algal blooms

The associations between environmental variables and bloom probability found in our study aligns with results from the literature. This is particularly true when we focus on the associations for water temperature and salinity, which is understandable given that algal taxa generally bloom in certain ranges of temperature and salinity. Water temperature was strongly associated with bloom probability, independently of training information type and algal taxon. Our results also showed strong associations for salinity for all taxa when collinearities were removed from the training information. The importance of water temperature and salinity for algal growth and their associations comply with multiple studies (Tango et al., 2005; Johnson, 2015; Fiorendino et al., 2020; Millette et al., 2023). For example, *Heterocapsa rotundata* blooms are predicted in colder temperatures and low salinity water, which concurs with Millette et al. (2023) who found high abundance of this taxon at optimal ranges of 0–10 °C and 7–12 salinity in the Chesapeake Bay. Our results also showed that *Dinophysis acuminata* blooms in colder and saltier waters, which agrees with Fiorendino et al. (2020) who found that this algal taxon blooms optimally in 12–24 °C and 22–34 salinity waters in a laboratory-controlled setting. The environmental information generated by the mechanistic model allowed us to also focus on variables that are difficult to observe directly such as vertical diffusivity and residual water flow velocity. We showed that vertical mixing and light extinction are associated with lower probability of blooms for most species. This is expected given that these factors decrease the ratio between euphotic and mixing depth and light climate, respectively. The residual water flow velocity was an important predictor for *Microcystis sp.* blooms. The reason is that this taxon mainly blooms in the upper reaches of the tributaries (Horemans

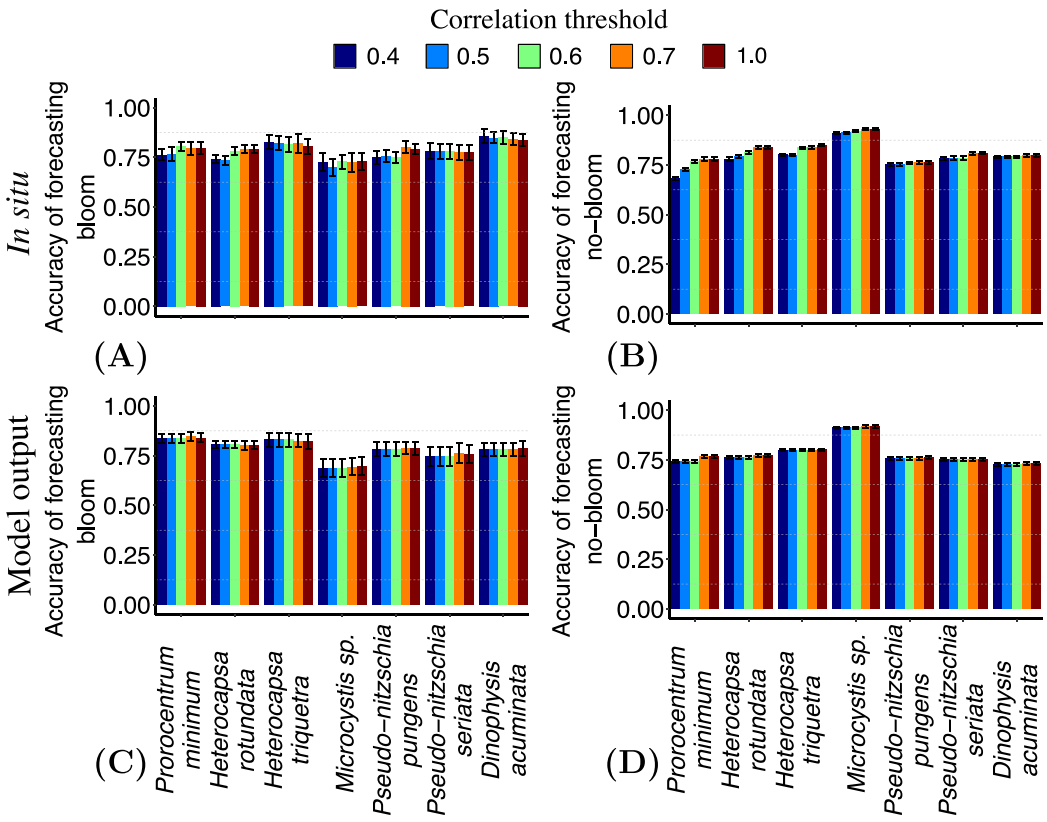


Fig. 4. Generalized linear model (GLM) accuracy of forecasting a bloom and no-bloom occurrence for the seven algal taxa examined using various correlation thresholds (colored symbols) in which the GLM is trained using environmental (A)–(B) *in situ* observations and (C)–(D) mechanistic model output. The colored vertical lines depict the standard deviations of the accuracies of the 50 GLMs that were constructed for each optimal set of environmental predictors.

et al., 2024) where variability in the water flow has an important impact on the hydrodynamics.

4.3. Generalizing our findings when using different species distribution model types

Various authors have studied the effect of correlative SDM types on species distribution forecasting (Elith, 2000; Pearson et al., 2006; Anderson et al., 2010; Valavi et al., 2022). To examine how results from a different SDM type would compare to our findings, we repeated our analysis using regression trees instead of GLMs (Supplementary Material; Figs. S5–S6). Most of our conclusions remain the same when using the nonlinear regression trees. The results show very similar patterns when comparing different taxa and information types. For example, *Microcystis sp.* shows the highest accuracy of forecasting a no-bloom occurrence, regardless of whether *in situ* observed or model-derived environmental fields were used to train the SDM, or the type of SDM applied. This similarity indicates that underlying observations and corresponding complex underlying species characteristics of these algal species have more impact on the model skill than biases in the mechanistic model output (Supplementary Material; Fig. S3) or, to a lesser extent, the SDM type used. This complies with Horemans et al. (2024) who demonstrated that although errors in SDM forecasts can be caused by using imperfect environmental fields derived from mechanistic models, these errors may be diminished by training SDMs using these same environmental fields. Horemans et al. (2024) used GLMs in their study; the fact that we found similar results with regression trees thus suggests that their conclusions are also likely true when using other types of SDMs. Our results also agree with those of Takolander et al. (2025) and Arsenault et al. (2025) who showed that species characteristics have a higher impact on the predictive skill than the SDM method applied and stressed the importance of observational response data quality, respectively. Similar to the GLM analysis, the model skill stays the same or slightly improves for increasing correlation threshold.

However, we also detect differences when comparing the analysis using GLM and regression trees. For example, we improve model skill after removing collinearity using regression trees for *Prorocentrum minimum*, *Heterocapsa triquetra*, and *Microcystis sp.* using *in situ* observations (p-value < 0.05 using a Welch two-tailed t-test). This improvement suggests that collinearity may reduce model skill when selecting the optimal variable combination based on Shapley values from regression trees rather than on direct model performance (i.e., Akaike Information Criterion) using GLMs. Another noticeable difference is that, although reducing collinearity can still influence the optimal combination of predictors, regression trees show less variability in these selections. Additionally, regression trees detect fewer strong associations, indicating higher variability in the (linear) association between environmental factors and bloom probability. This result complies with expectations because regression trees capture nonlinear patterns while generalized linear models assume a linear relationship between (logit-transformed) bloom probability and environmental conditions [Eq. (3)]. Despite these relatively minor differences when comparing GLMs and regression trees, overall the results that support our main findings are similar.

4.4. The niche concept and its implications for statistical forecasting of estuarine algal blooms

Interpreting the associations between environmental variables and algal blooms is challenging and is affected by other factors beside the training information type. When interpreting associations between environmental conditions and algal bloom occurrences, we typically think in terms of the Grinnellian Niche, i.e., species arise whenever and wherever the environmental conditions are suitable and nowhere else.

Put another way, we presume that environmental variables drive algal blooms and whenever the environmental variables are suitable, we expect a bloom to occur. However, besides the niche of a species being driven by the environmental requirements, the environment may also be altered by the population of an individual taxon (Leibold, 1995). For example, most taxa in our study demonstrated a positive relationship between TON and bloom probability (Fig. 3). This relationship may result because algae produce organic material, which increases the TON concentration (*impact of the species on their environment*) and not because TON *caused* an algal bloom to appear (*driven by the environmental requirements*). Similarly, algae produce oxygen and consume CO_2, which may explain the negative and positive relationship between most algal blooms and AOU and pH, respectively.

Also, the co-existence of competing algal taxa may result in the absence of an otherwise expected algal bloom given favorable environmental conditions (cf., Hutchinsonian Realized Niche). For example, the associations between environmental variables and bloom probability are very similar for *Prorocentrum minimum*, *Heterocapsa rotundata*, and *Heterocapsa triquetra* (Fig. 3). From a Grinnellian Niche perspective, we may interpret these results as if these taxa are competing for the same resources. This has important consequences when forecasting algal blooms; based on suitable environmental conditions, we may incorrectly forecast the presence of a specific algal taxon because of the existence of a dominant competitor. Another way of looking at the overlapping associations is that these taxa prefer to co-exist (cf., symbiosis). This co-existence may inflate statistical parameter variability similar to the impact of collinearity between environmental variables. Additionally, the overlapping associations found for *Prorocentrum minimum*, *Heterocapsa rotundata*, and *Heterocapsa triquetra* may be explained by these taxa emigrating from the same external source habitat (cf., Source-Sink Dynamics). The Source-Sink Dynamics niche may be particularly important in estuarine systems that are driven by water currents applied to algal species, which are often passive and not actively migrating (with the exception of vertical migration). However, overall, our results do not show a strong association between algal bloom probability and water currents when compared to other variables such as temperature and salinity (cf., flow in Fig. 3), suggesting that the influx of algae into the system (cf., source) is not a main contributor for bloom occurrence, at least for our application.

Finally, another crucial assumption commonly made when applying statistical methods that may have important consequences on interpretation of the correlative SDM results is that the response of a bloom to a change in environmental conditions is essentially instantaneous (i.e., within one day) (e.g., Guisan and Thuiller, 2005). This assumption of (pseudo-) equilibrium may not be valid as it may take multiple days, weeks, or even months for an algal bloom to develop. Cyst dynamics (Li et al., 2021) and grazing (causing limit cycles; Steele and Henderson, 1992), for example, may result in multi-month variability, hindering the equilibrium condition. The assumption of equilibrium is particularly important in the Dispersal Limitation niche because it may take time for an algal bloom to recover from an extinction event.

All these factors may hinder the detection of the importance of a specific environmental variable to bloom occurrence using statistical approaches, showing that interpreting these results and their application to forecasting should be taken with care.

5. Summary and conclusions

Correlative species distribution models (SDMs) have been broadly used to predict species' distributions in time and space by relating environmental conditions to organism presence without the requirement of *a priori* insights into species dynamics. Interpreting these relationships, however, should be taken with care given their biological complexity. Our results highlight these challenges and suggest that the selection of the essential environmental predictors and their association with species presence not only rely on the intrinsic biological characteristics of the species dynamics but also on the training information type (i.e., level of collinearity and *in situ* observations versus mechanistic model simulations) used in SDM construction. Although future research is needed to validate our conclusions when applied to other, non-aquatic taxa, our results are crucial for further advances in developing SDMs and interpreting their results. Additionally, we have elucidated the key environmental predictors and their association with the likelihood of blooms caused by seven algal species in the Chesapeake Bay. This can help optimize monitoring efforts, such as those performed by the Chesapeake Bay Program, by revealing which *in situ* observations are vital to accurately measure and forecast blooms of these estuarine algae, many of which are harmful.

CRediT authorship contribution statement

Dante M.L. Horemans: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jennifer C. Lin:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Marjorie A.M. Friedrichs:** Writing – review & editing, Project administration, Investigation, Funding acquisition, Conceptualization. **Pierre St-Laurent:** Writing – review & editing, Resources, Investigation, Funding acquisition, Conceptualization. **Raleigh R. Hood:** Writing – review & editing, Investigation, Funding acquisition, Conceptualization. **Christopher W. Brown:** Writing – review & editing, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ecoinf.2025.103225.

Data availability

The *in situ* observations are publicly available through the Chesapeake Bay Program data hub (https://www.chesapeakebay.net/what/data; Chesapeake Bay Program, 2025). ROMS-ECB model code and outputs are archived at https://doi.org/10.25773/q2kh-rd09. The specific environmental model output and *in situ* observations that were used in this contribution are accessible via https://doi.org/10.5281/zenodo.15466597. We used open-source R software (version 4.2.1) to develop the generalized linear models (*glm* function from R-base *stats* package https://www.r-project.org/; R Core Team, 2022), regression trees (*rpart* function from *rpart* package https://doi.org/10.32614/

CRAN.package.rpart; Therneau and Atkinson, 2022), and corresponding Shapley values (*explain* function from *fastshap* package https://doi.org/10.32614/CRAN.package.fastshap; Greenwell, 2024). All relevant R-package versions and corresponding code are archived and publicly accessible via https://cran.r-project.org/.

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